MODULE 01 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Event-Driven Architecture and Delivery Guarantees

At-least-once is the default everywhere. Ordering, deduplication and routing are choices you make.

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Every AWS messaging service can deliver a message **more than once**. Exactly once is something your consumer does.

What each service actually promises

SERVICE	DELIVERY	ORDERING
SQS standard	At least once	Best effort only
SQS FIFO	Deduplicated for 5 minutes	Strict, per message group ID
SNS standard	At least once, retries then drops	None
SNS FIFO	Deduplicated for 5 minutes	Strict; queues as subscribers
EventBridge	At least once, 24 h / 185 retries	None
Kinesis Data Streams	At least once	Strict, per shard, by partition key

ON THE EXAM

Ordering on AWS is always scoped: per message group or per partition key. Nothing gives you global order cheaply.

Where do you remove duplicates?

AT THE BROKER

FIFO deduplication

- Deduplication ID or content hash
- Five-minute window only
- Catches producer retries
- Does not stop redelivery after a consumer fails

IN THE CONSUMER

Idempotent processing

- Key taken from the business event
- Conditional write before the side effect
- Survives retries, replays and redrives
- Works on any broker

THE TIEBREAKER

Broker deduplication stops duplicate sends. Only an idempotent consumer stops duplicate effects.

Building an idempotent consumer

- **Key from the event,** not the message: order ID, payment ID, event ID.
- **Conditional write first.** DynamoDB put with attribute_not_exists on the key.
- **Expire keys with TTL** once replays can no longer arrive.
- **Powertools for AWS Lambda** ships an idempotency utility built on this pattern.
- **Not the SQS message ID.** A producer retry creates a new one.

> ON THE EXAM
>
> The idempotency key must identify the business action. A broker-generated ID changes on every resend.

Ordered processing: FIFO queue or Kinesis?

CHOOSE SQS FIFO WHEN

Ordered work queue

- One consumer group does the work
- Order per entity via message group ID
- Messages are deleted once processed
- Per-message retry and dead-letter queue

CHOOSE KINESIS WHEN

Ordered, replayable log

- Several consumers read independently
- Order per entity via partition key
- Data must be replayed from a point in time
- Very high, sustained throughput

THE TIEBREAKER

Several independent readers, or a need to reprocess history, means a stream. One worker pool means a queue.

Ordering costs throughput

- **One message group is one lane.** Messages in it are processed strictly in sequence.
- FIFO default: **300 TPS per partition**, per API action, 3,000 with batches of 10.
- **High throughput mode** reaches 70,000 TPS in the largest Regions.
- **A constant group ID** serialises the whole queue, whatever the mode.
- **Ask what must be ordered:** per shipment, per account. Never everything.

> ON THE EXAM
>
> Use the entity ID as the message group ID. Order where the business needs it, parallel everywhere else.

Where failed messages go

SOURCE	FAILURE HANDLING
SQS	Redrive policy: maxReceiveCount, then a dead-letter queue
SNS subscription	Subscription-level dead-letter queue
EventBridge target	Retry policy per target, then an SQS dead-letter queue
Lambda async invoke	On-failure destination: SQS, SNS, EventBridge or Lambda
Kinesis to Lambda	Bisect batch, max retries, on-failure destination

ON THE EXAM

Without a dead-letter queue or destination, EventBridge and SNS eventually drop the event. Silence is the default.

Cross-account and cross-Region event routing

01
Produce
Workload account puts events on its own bus.

02
Forward
A rule sends matching events to a central bus.

03
Govern
Central bus policy allows the organisation only.

04
Route
Central rules deliver to targets in other accounts.

05
Replicate
Rules forward to a bus in another Region.

ON THE EXAM

Skill 1.1.5 is about event buses and forwarding. The central bus is where policy, archive and replay live.

Deliver to a target, or to their bus?

DIRECT CROSS-ACCOUNT TARGET

Since January 2025

- SQS, Lambda, SNS, Kinesis Data Streams, API Gateway
- Target owner adds a resource policy
- No EventBridge resources in the receiving account
- Same Region only

FORWARD TO THEIR EVENT BUS

Bus to bus

- Receiving team owns its own rules
- One delivery fans out to many targets
- Required for any cross-Region hop
- Rule needs an IAM role to put events

THE TIEBREAKER

Crossing a Region means a bus. Same Region, and the receiver wants nothing to manage, means a direct target.

Securing and operating the central bus

- **aws:PrincipalOrgID** in the bus policy admits the whole organisation in one statement.
- Forwarded events keep their original **account** and **source** fields.
- **Watch for loops** when a forwarded event matches a rule that forwards it back.
- **Archive and replay** on the central bus recovers events after a consumer bug.
- **Global endpoints** fail ingestion over to a secondary Region.

> ON THE EXAM
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> One condition key, aws:PrincipalOrgID, replaces a bus policy that lists every account by number.

Forwarding rule or global endpoint?

CROSS-REGION RULE

Copy events onward

- Runs inside the source Region
- Producers keep publishing to one Region
- Useless if that Region's EventBridge is impaired
- Good for aggregation and analytics

GLOBAL ENDPOINT

Fail ingestion over

- Route 53 health check decides the active Region
- Same-named custom bus in both Regions
- Replication needed for automatic failback
- RTO and RPO of 360 seconds, 420 at most

THE TIEBREAKER

If the requirement is that producers keep publishing when a Region fails, the rule is the wrong layer.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Charged twice, processed twice	Idempotent consumer, not a new broker
In order for each customer	Message group ID or partition key
Several teams, reprocess history	A stream, not a queue
Events silently missing	A dead-letter queue that was never configured
Receiving team manages nothing	Direct cross-account target
Keep publishing if a Region fails	EventBridge global endpoint

ON THE EXAM

Each phrase removes options before you read them. Find it in the scenario first.

> Meridian Freight's payments service is a Lambda function triggered by an SQS standard queue. Each message charges a customer's card through a third-party API. Function duration is under two seconds; the visibility timeout is five minutes.
>
> A few customers each week are charged twice for the same booking.

What should the solutions architect do?

A Increase the visibility timeout to 15 minutes

B Replace the queue with an SQS FIFO queue using content-based deduplication

C Record the booking ID in DynamoDB with a conditional write before charging, and skip the charge if the write fails

D Replace SQS with an SNS topic that invokes the function directly

◆ PAUSE AND DECIDE

What should the solutions architect do?

A. Longer visibility timeout

The function already finishes far inside five minutes. The timeout is not what causes the duplicates.

B. FIFO with deduplication

Removes duplicate sends within five minutes. A crash after charging but before delete still redelivers.

C. Idempotency key in DynamoDB

Makes the side effect safe to repeat, whatever caused the redelivery.

D. SNS invoking Lambda

Still at least once, and loses the queue's buffering and retry control.

THE SIGNAL

Duration well inside the visibility timeout rules out A. Duplicated side effects point at the consumer, not the broker.

> Meridian Freight processes shipment status events through an SQS FIFO queue. About 40,000 shipments are active at once, and each shipment's events must be applied in the order they occurred.
>
> Every message uses the message group ID status-updates. The backlog grows every afternoon, while the consuming Lambda function shows low concurrency.

What change resolves the backlog while preserving the required ordering?

A. Use the shipment ID as the message group ID

B. Enable high throughput mode on the existing queue

C. Raise the Lambda function's reserved concurrency

D. Move to an SQS standard queue and sort events by timestamp in the consumer

◆ PAUSE AND DECIDE

What change resolves the backlog while preserving the required ordering?

A Shipment ID as group ID

Order is kept per shipment, and 40,000 groups can be processed in parallel.

B High throughput mode

Scales by spreading groups. With a single group there is nothing to spread.

C More reserved concurrency

One message group delivers to one consumer at a time. Extra concurrency sits idle.

D Standard queue and sorting

A consumer cannot sort events that have not arrived yet. Ordering is lost.

THE SIGNAL

Low concurrency with a growing FIFO backlog means too few message groups. The requirement was per shipment, not global.

> 2,000 Meridian Freight trucks each send 10 telemetry readings per second. Three teams consume the readings independently: routing, maintenance and data science. Readings from each truck must be processed in order.
>
> After retraining a model, data science needs to reprocess the previous seven days of readings.

Which design meets these requirements?

A. An EventBridge custom bus with an archive, replaying events when needed

B. An SNS standard topic fanning out to three SQS standard queues

C. One SQS FIFO queue per team, with the producer writing to all three

D. Kinesis Data Streams, partitioned by truck ID, with retention extended to seven days

◆ PAUSE AND DECIDE

Which design meets these requirements?

A EventBridge with archive

Replay exists, but EventBridge guarantees no ordering at all.

B SNS standard fan-out

Gives each team its own copy, but neither standard SNS nor standard SQS keeps order.

C Three FIFO queues

Couples the producer to every team, and nothing keeps readings after they are deleted.

D Kinesis by truck ID, seven days

Ordered per truck, independent consumers, and replay from a point in time.

THE SIGNAL

Independent consumers plus reprocessing history is the signature of a stream. A queue forgets a message once it is deleted.

> Meridian Freight's platform account publishes customer-deleted events to a custom EventBridge bus in eu-west-1. The data protection team owns an SQS queue in a separate account in the same Region and must receive only these events.
>
> The data protection team does not want to create or manage any EventBridge resources.

Which solution meets the requirements with the LEAST operational overhead?

A. A Lambda function in the platform account that assumes a role and calls SendMessage

B. A rule in the platform account targeting the queue directly, with a queue policy allowing the rule

C. A rule targeting the data protection account's default bus, plus a rule there targeting the queue

D. An SNS topic in the platform account with the queue subscribed to it

◆ PAUSE AND DECIDE

Which solution meets the requirements with the LEAST operational overhead?

A Lambda with SendMessage

Works, but it is custom code doing what a rule target now does natively.

B Direct cross-account target

One rule, one queue policy. Nothing EventBridge-related in the receiving account.

C Bus to bus

Puts a rule in the data protection account, which the team explicitly does not want.

D SNS in between

Adds a second service and its own policy for no benefit over a direct target.

THE SIGNAL

Same Region, and the receiver wants nothing to manage. Since January 2025 that is a direct cross-account target.

> Meridian Freight runs a warm standby of its booking platform in eu-central-1. Booking events published to a custom bus in eu-west-1 must also be processed by a Lambda function in eu-central-1, so the standby's database stays current.
>
> The primary Region is healthy; this is about keeping the standby in step.

How should events reach the function in eu-central-1?

A. A rule in eu-west-1 targeting an event bus in eu-central-1, with a rule there invoking the function

B. A rule in eu-west-1 targeting the Lambda function in eu-central-1 directly

C. A rule in eu-west-1 targeting an SQS queue in eu-central-1 that triggers the function

D. An EventBridge global endpoint with event replication enabled

◆ PAUSE AND DECIDE

How should events reach the function in eu-central-1?

A Bus to bus across Regions

The supported cross-Region pattern: the target of a cross-Region rule is an event bus.

B Lambda in another Region

Cross-Region rule targets must be event buses, not functions.

C SQS in another Region

Same problem. Direct targets work across accounts, not across Regions.

D Global endpoint

Solves a different problem: failing ingestion over when a Region is impaired.

THE SIGNAL

Cross-account direct targets do not extend across Regions. Any Region hop goes through an event bus.

Meridian Freight's order services publish to a custom EventBridge bus in us-east-1. A rule already forwards every event to a bus of the same name in us-west-2.

During a regional EventBridge impairment, producers received errors and orders were lost. The business wants publishing to continue automatically in a future impairment, and to return to us-east-1 automatically afterwards.

What should the solutions architect implement?

A Add a dead-letter queue to the existing forwarding rule

B Have every producer publish each event to both Regions

C Put a Route 53 failover record in front of the EventBridge API endpoints

D An EventBridge global endpoint with a Route 53 health check and event replication enabled

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A DLQ on the forwarding rule

The events never reached the bus. A rule's dead-letter queue cannot catch a failed PutEvents.

B Publish to both Regions

Duplicates every event and pushes failover logic into every producer.

C Route 53 in front of the API

You cannot place your own DNS record in front of AWS service endpoints.

D Global endpoint with replication

Fails ingestion over, and replication is what allows automatic failback.

THE SIGNAL

The forwarding rule existed and still lost orders. Failures at ingestion need a layer in front of the Region.

What to carry out of this module

- **At least once everywhere.** Duplicate effects are fixed in the consumer, with a business key.
- **Ordering is always scoped:** message group ID or partition key, never global.
- **Stream for many readers and replay,** queue for one worker pool.
- **Direct cross-account targets** since January 2025. Cross-Region always needs a bus.
- **Global endpoints** protect ingestion. Forwarding rules cannot.

> ON THE EXAM
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> Stale fact to watch: SQS messages now go up to 1 MiB, not the 256 KB older material teaches.